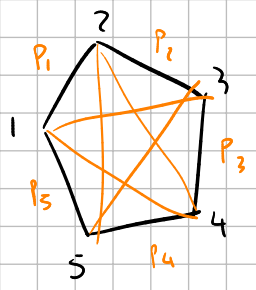


x_{13}

x_{14}

x_{24}

x_{25}

x_{35}

$$c_{ij} = -s_{ij} = -(p_i + p_j)^2$$

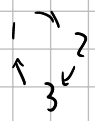
$$x_{13} = (p_1 + p_2)^2$$

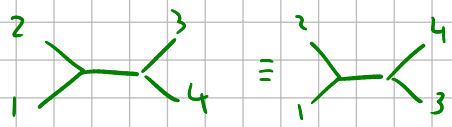
↳ planar

$$x_{25} = (p_2 + p_3 + p_4)^2$$

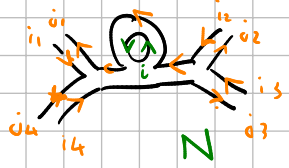
$c_{12} = x_{13}$ planar

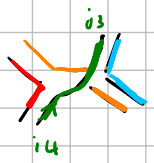
$c_{13} = (p_1 + p_3)^2$ not planar if $n > 3$



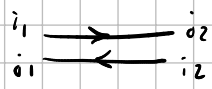


$$\langle \phi^{i_1}_{j_1}(p_1)\, \phi^{i_2}_{j_2}(p_2) \cdots \rangle$$

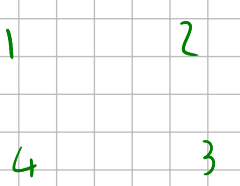


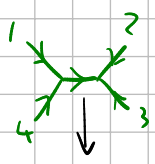


$$\mathcal{L} = \frac{1}{2}\, \mathrm{tr}(\underbrace{\partial_\mu \phi\, \partial^\mu \phi}_{\text{↳ gives propagator}}) + V(\phi)$$

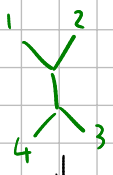


$$\sum_{\substack{i_1 j_1 \\ i_2 j_2}} \partial_\mu \phi^{i_1}_{j_1}\, \partial^\mu \phi^{i_2}_{j_2}\, \delta^{j_1}_{i_2}\, \delta^{i_1}_{j_2}$$

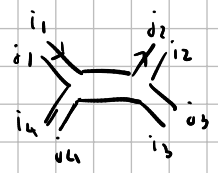


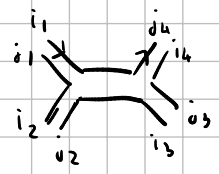


$$\frac{1}{(p_1+p_4)^2}$$

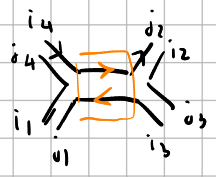


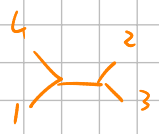
$$\frac{1}{(p_1+p_2)^2}$$



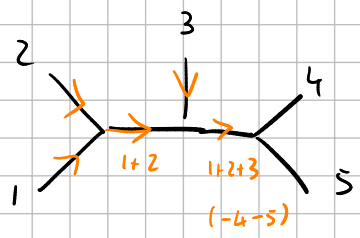


$$\frac{1}{(P_1+P_4)^2}\,\delta^{i_1}_{j_2}\,\delta^{i_4}_{j_3}\cdots \qquad \frac{1}{(P_1+P_2)^2}\,\delta^{i_1}_{j_4}$$





$$\frac{1}{(P_1+P_4)^2}\,\delta^{i_1}_{i_4}$$

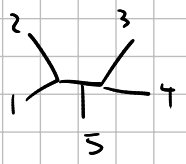


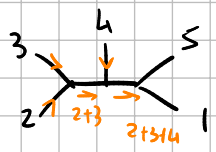
$$\frac{1}{x_{13}} \quad \frac{1}{x_{14}}$$

$$s_{12} \qquad s_{123}$$

$$A_5(1,2\,3\,4\,5) = \frac{1}{x_{13}\,x_{14}} + \text{cyclic}$$

$$\frac{1}{x_{24}\,x_{25}} + \frac{1}{x_{35}\,x_{31}} + \cdots$$





$$x_{24} \qquad x_{25}$$

$$\frac{1}{s_{12}} \quad \frac{1}{s_{12}}$$

$P \cdot P$

$$P^2 = 0 \quad \rightarrow \quad E^2 = \vec{p}^{\,2}$$

energy of a photon

Minkowski space: $\quad P_\mu P^\mu = 0 \qquad P_t P_t - P_x P_x - P_y P_y - P_z P_z = 0$

$P_t = E$ energy

$\vec{p} = (P_x\, P_y\, P_z)$ momentum

$$A_4 \propto \frac{1}{s_{12}} + \frac{1}{s_{13}} + \frac{1}{s_{14}} =$$

$$\frac{1}{s_{13}} \quad A_{left}\ A_{right}$$